

SUMMARY

Machine Learning Engineer focused on building production-grade AI systems, including multi-agent architectures, NLP pipelines, and LLM-integrated applications. Experienced in designing end-to-end ML solutions with backend APIs, real-time dashboards, and explainable AI.

EXPERIENCE

Project Intern, Zephyra

11/2024 – 1/2025

- Built NLP-based chatbot using intent classification and hybrid rule-based + ML pipeline
- Designed backend workflows for real-time query handling and response generation
- Contributed to deployment-ready AI system development

ML Intern, Unified Mentor

6/2025 – 8/2025

- Developed ML models for classification and recommendation systems on real-world datasets
- Built end-to-end pipelines including preprocessing, feature engineering, and evaluation
- Improved model performance using tuning and validation techniques

ML Intern, CodexIntern

7/2025 – 8/2025

- Built ML models including spam detection and house price prediction systems
- Applied feature engineering and model evaluation techniques
- Developed end-to-end ML workflows for practical applications

PROJECT

Cognitive SOC Framework (Multi-Agent AI Security System) : <https://multi-agent-cognitive-soc-framework.onrender.com>

- Engineered a multi-agent AI architecture (14 agents) for real-time cybersecurity threat detection and automated response
- Developed anomaly detection pipeline using Random Forest and Isolation Forest, improving detection of abnormal system behavior
- Built a decision intelligence layer with confidence scoring and risk prioritization
- Integrated Explainable AI techniques (SHAP, LIME, LLM-based reasoning) for transparent model predictions
- Designed scalable backend services using FastAPI for real-time data ingestion and API interaction

AI-Powered Student Guidance & Analytics System : <https://mitigating-educational-uncertainty-xwrr.onrender.com>

- Built a full-stack AI platform integrating 6 machine learning models for personalized stream and course recommendations
- Developed LLM-based assistant using LangChain for interactive career guidance
- Designed and implemented 50+ REST APIs using FastAPI for system communication and data flow
- Implemented predictive models for dropout risk detection and early intervention
- Created unified dashboard for analytics, recommendations, and performance tracking

EDUCATION

Bachelor of Technology (B.Tech), Computer Science
Oriental Institute of Science and Technology, Bhopal
CGPA: 7.17 | Expected Graduation: 2027

ACHIEVEMENTS

Winner – QubitX 2025 National Hackathon, L Bajaj Group of Institutions, Mathura

5/2025

- Secured 1st position among 400+ teams nationwide; excelled across multiple rounds including idea validation, mentoring, and final evaluation

First Runner-Up – Research Paper Publication, Kirori Mal College, Delhi University (Pareto Time'25)

4/2025

- Ranked among Top 50 finalists out of 2000+ participants; developed data-driven policy solutions and presented to domain experts

Second Runner-Up – Policy Hackathon, IIT Kanpur (Policy Conclave '26)

1/2026

- Research paper titled “Quantum Leaps in Industry: The Economic Alchemy of Artificial Intelligence” was selected, presented, and officially published.

Solved 100+ LeetCode problems (DSA focused, actively improving)

RESEARCH PUBLICATIONS

- Published in International Journal of Scientific Research in Technology & Management (IJSRTM), Jan 2026
- “Reducing Decision Uncertainty in AI-Based Student Career Guidance using Hybrid ML & LLM Framework”
- Proposed hybrid architecture combining machine learning models and LLMs for improved decision-making

SKILLS

Languages: Python, C++, SQL
ML/AI: Scikit-learn, TensorFlow, PyTorch, OpenCV, NLP (SpaCy), SHAP, LIME
LLM & Systems: LangChain, LangGraph, Mistral AI, OpenAI, RAG, Prompt Engineering
Backend: FastAPI, Flask, REST APIs
Tools: Git, MongoDB, Docker, Jupyter, VS Code

CERTIFICATES

- Oracle Cloud Infrastructure 2025 – AI Foundations Associate
- NPTEL – Introduction to Machine Learning
- AI/ML Industrial Training Program – Pregrad
- AI/ML Certification – Rinex